MATTHEW EDGAR

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EDUCATION

University of California, Berkeley **May 2026**
Bachelor of Science in Chemical Biology – Concentration in Computational Chemistry

San Diego Mesa College **May 2024**
Associate of Science in Chemistry
Associate of Arts in Math and Pre-Engineering Math
Graduated with Honors

Relevant Coursework:

- **Programming:** C/C++ Programming, Python Programming for Engineers, Computational Chemistry, Numerical Analysis
- **Mathematics:** Discrete Mathematics, Mathematical Tools for the Physical Sciences, Linear Algebra, Differential Equations

EXPERIENCE

Williams Research Lab – University of California, Berkeley **January 2025 – May 2026**
Undergraduate Researcher
Berkeley, CA

- Designed and assembled a nanoelectrospray apparatus, including custom circuitry and resistive components.
- Conducted 200 nanoelectrospray ionization experiments and developed Python scripts to collect and analyze their data.
- First author on a pending manuscript based on this research.

LifeWest Ambulance Company **May 2024 – June 2025**
Emergency Medical Technician
Oakland, CA

- Transported injured and critically ill patients to and from medical facilities across the San Francisco Bay Area.

United States Navy **June 2017 – June 2021**
Machinist's Mate (E-5)
San Diego, CA

- Active-duty enlisted sailor in the engineering departments of two ships; the USS Ardent and the USS Makin Island.
- Operated and maintained shipboard engineering systems, including refrigeration, diesel engines, and reverse-osmosis systems.
- Maintained fuel and oil systems, weapons systems, and kept proper watchstanding logs for quality assurance.

PROJECTS

Nanoelectrospray Research
Williams Research Lab – University of California, Berkeley – Python

- Developed a Python script that extracts the RGB values of specific pixels in an MP4 and records them in a CSV.
- A separate script extracts the RGB values from the CSV, averages, and plots them as a function of time.
- By comparing these plots to RGB values at known acidities, pH values can be estimated at different time points and the buffering capacities of different chemical agents can be deduced. This is useful for applications in mass spectrometry.

Monte Carlo Sampler using Egg Box Potential
Python

- Created a Python script that uses the Monte Carlo method to estimate the egg box energy integral of a particle.
- This project further applies different forces to the particle and simulates the positions of two interacting particles.

Weight Loss iOS Application
Swift, Xcode

- Independently developing an iOS application in Xcode that assists consumers with weight loss and is linked to social media apps.
- Implementing user interface and third-party integration with other applications.

Personal Website
HTML, Jekyll, CSS, GitHub Pages

- Developed a personal portfolio website to display my research, professional experience, and software projects.
- Built and deployed this site using GitHub Pages and Jekyll.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB
Scientific Computing: NumPy, Matplotlib, Monte Carlo Methods, Numerical Analysis and Estimation